Kai Li

Data Science, Applied Machine Learning, and Reinforcement Learning

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SUMMARY

Statistics Ph.D. candidate with peer-reviewed research in reinforcement learning, optimization, simulation, and statistical machine learning. Builds reproducible computational studies in Python and R, including custom environments, controlled model comparisons, replicated experiments, held-out evaluation, and statistical analysis. First author of a journal article and co-first author of an ICML workshop paper; seeking 2027 data science, applied scientist, machine learning, or reinforcement learning roles.

TECHNICAL SKILLS

Programming: Python, R, SQL. **Libraries and frameworks:** scikit-learn, TensorFlow/Keras, Gymnasium, Stable-Baselines3, mpi4py.
ML and RL: supervised learning, neural networks, Monte Carlo control, SARSA, Q-learning, deep Q-networks, proximal policy optimization, Markov decision processes.
Statistics and computing: regression, experimental design, time series, simulation, Monte Carlo methods, model evaluation, optimization, simulated annealing, data visualization, bootstrap methods, parallel and high-performance computing, MPI.
Tools: Slurm, GNU Parallel, Git, GitHub, Jupyter, RStudio, Visual Studio Code, LaTeX.

SELECTED RESEARCH AND APPLIED MACHINE LEARNING EXPERIENCE

Reinforcement Learning Researcher — Blackjack Strategy **September 2025–Present**
Collaborative research project, Stony Brook University *Stony Brook, NY*

- **Publication study (March 2026–Present):** Compare six tabular and deep reinforcement learning algorithms: on-policy Monte Carlo, off-policy Monte Carlo, SARSA, Q-learning, deep Q-networks, and proximal policy optimization.
- Study algorithm behavior as the legal action space expands from Hit/Stand to settings that also include Double and Split, using a common Gymnasium-compatible simulator with state-dependent action handling.
- Use replicated training, held-out seeds, learning curves, measures of policy agreement, and matched decision-level comparisons; preparing a manuscript as a co-first author.
- **Pilot study (September–December 2025):** Compared Monte Carlo control and proximal policy optimization and identified confounding from action-space differences, split-hand credit assignment, reward variance, and deck assumptions.

Researcher and First Author — Cumulative Tic-Tac-Toe **July 2024–March 2026**
Department of Applied Mathematics and Statistics, Stony Brook University *Stony Brook, NY*

- Formulated a cumulative-scoring game as a Markov decision process and built a temporal-difference learning framework for empirical policy evaluation.
- Combined exhaustive game analysis and combinatorial game theory to establish that the game is a draw under optimal play.
- Designed a domain-informed evaluation function that reduced average training episodes by approximately 23.1% versus random initialization.
- Published the completed work as a first-author article in *Stats*; maintained a public research code repository tied to the published computational record.

Researcher and Co-First Author — RL-QESA **December 2024–July 2025**
Collaborative research project, Stony Brook University *Stony Brook, NY*

- Modeled temperature control for simulated annealing as a block-level Markov decision process in which cooling is gated by a quasi-equilibrium condition.
- Used a transformer-based policy trained with proximal policy optimization to adapt cooling decisions from block-level energy and acceptance statistics.
- Helped establish conditions retaining classical convergence guarantees for simulated annealing and evaluated performance on Rosenbrock and Lennard–Jones optimization tasks against classical and neural baselines.
- Published the work at the AI for Math Workshop at ICML 2025 as a co-first author.

Graduate Research Assistant **May 2025–August 2025**

Institute for Advanced Computational Science, Stony Brook University *Stony Brook, NY*

- Assessed reinforcement learning methods for climate model parameterization and multi-agent adaptation to sea-level rise through technical literature review and mathematical formulation.
- Compared eight model-free algorithms on sample efficiency, stability, hyperparameter sensitivity, and reproducibility; translated a published adaptation model into Markov decision process components and Bellman equations.
- Produced two technical presentations and an implementation-oriented outline for a future multi-agent advantage actor–critic simulation study.

SELECTED DATA SCIENCE PROJECTS

Scalable Statistical Computing on SeaWulf **September 2026–Present**

- Compared model-based linear regression, a standard bootstrap with 5,000 resamples, and the Bag of Little Bootstraps on 500,000 observations; all three methods produced closely aligned estimates and identical conclusions about which coefficients excluded zero.
- Parallelized the resampling workflow on SeaWulf using Slurm and GNU Parallel across five CPU processes, with controlled numerical-library threading and fixed random seeds for reproducibility.
- Implemented MapReduce-style PageRank using Python and mpi4py across six MPI processes on SeaWulf, with chunked graph input, collective reductions, and dangling-page handling; verified repeatable top-ten results after four prescribed iterations.

Time Series Forecasting of Store Sales **February–April 2022**

- Implemented ARIMA, recurrent neural network, LSTM, and GRU forecasting workflows in R and Keras and compared predictive performance using root mean squared error.

Understanding Flight Delays **August–December 2021**

- Prepared 2012–2019 airline data and compared ridge regression, multilayer perceptrons, and k-nearest neighbors; used statistical tests to assess model performance and analyzed seasonal and airline-specific delay patterns.

ADDITIONAL EXPERIENCE

Graduate Instructor and Teaching Assistant **August 2022–Present**

Department of Applied Mathematics and Statistics, Stony Brook University *Stony Brook, NY*

- Taught or supported nine undergraduate and graduate sections in statistical computing, mathematical statistics, time series, and statistical learning, serving 320 students; earned five teaching distinctions, including two Teachers Rated Excellent Educators by their Students (TREES) recognitions.

EDUCATION

Stony Brook University, Stony Brook, NY **Expected May 2027**

Ph.D. in Applied Mathematics and Statistics, Statistics Track; GPA: 4.00/4.00

Stony Brook University, Stony Brook, NY **December 2022**

M.S. in Applied Mathematics and Statistics; GPA: 3.95/4.00. Completed requirements for both Statistics and Operations Research tracks.

Stony Brook University, Stony Brook, NY **2022**

Advanced Graduate Certificates in Operations Research and Data and Computational Science

The Ohio State University, Columbus, OH **May 2020**

B.S. in Mathematics, Theoretical Track; minors in Computer Information Science and Economics

SELECTED RESEARCH OUTPUTS

- **Li, K.** and Zhu, W. "Combinatorial Game Theory and Reinforcement Learning in Cumulative Tic-Tac-Toe via Evaluation Functions." *Stats*, 9(2), Article 28, 2026. doi:10.3390/stats9020028.
- Xu, R., **Li, K.**, Wang, H., Kementzidis, G., Zhu, W., and Deng, Y. "RL-QESA: Reinforcement-Learning Quasi-Equilibrium Simulated Annealing." *2nd AI for Math Workshop at ICML 2025*, 2025. OpenReview. Co-first author.